Sergio Adan Bermudez, PhD

AI Engineer

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PROFILE

AI engineer with a PhD in sensing and measurement from the University of Edinburgh. I build large language model (LLM) systems that run on real hardware, and I measure every change: retrieval-augmented generation (RAG) with checked citations, AI agents whose safety limits live in code, evaluation harnesses with held-out test sets, and inference performance down to the GPU memory path. Nine years in optical sensing and measurement, in research and in industry, including two years in customer-facing technical roles.

PROJECTS

2026. Built with AI coding agents (Claude Code) writing most of the code, designed, directed and reviewed by me; every repository says so. The numbers are measured and each links to its source.

Argus: a local agent system, and a llama.cpp latency fix

- Runs my assistant on a local 35B mixture-of-experts model (Qwen3.6-35B-A3B, llama.cpp on Vulkan, AMD integrated GPU) inside the open-source Odysseus hub, with an MCP memory server I wrote (seven tools, path confinement, no deletes), a bridge that hands hard jobs to Claude Code, voice, and a supervisor that restarts failed services.
- Chose every setting by evaluation: 13 real tasks with deterministic checks; thinking off scored 13/13 at a 2.6 s median against 10/13 at 19.8 s with thinking on.
- Traced a 3-second pause before every spoken answer to the Vulkan backend reading uncached GPU memory (880 MB in 7.7 s). One condition cut a state snapshot from 0.61 s to 0.03 s and the first spoken word to about 0.7 s, with identical output. The maintainer had merged the same fix two weeks earlier, so it is an independent diagnosis, not a contribution.

sergioadan.dev/argus · sergioadan.dev/work/llama-cpp-stall

LabDesk: a documentation assistant, run as a client engagement

- Took a simulated client brief (PyVISA instrument-control support) to handover: testable requirements, six architecture decision records, an evaluation spec, a FastAPI service with health and metrics endpoints, and a runbook.
- BM25 retrieval over 98 documentation sections; every sentence cites its source and citations are checked against what the model was given; one fixed refusal sentence; a read-only power-meter tool.
- Held-out set committed to git before the first run: 26/31 correct (target 0.85, missed by one answer and reported rather than tuned), 27/31 correct citations, 8/8 off-topic questions refused, p95 latency 4.8 s on a local model.

github.com/sadanbermudez/labdesk-copilot

photonics-lab-mcp: an AI agent runs a lab experiment

- An MCP server with ten tools over a simulated optics bench from my PhD experiment, with the safety limits in code: a power cap, an arm-then-emit interlock, permission tiers, a safe state on any fault, and an append-only notebook that records refusals. 22 tests in CI.
- With the answer hidden in the server, a local model measured a photon order of 2.004 ± 0.03, and 1.003 ± 0.002 on a one-photon control. Its closing physics was wrong in both runs, and I published that.

sergioadan.dev/work/lab-agent · github.com/sadanbermudez/photonics-lab-mcp

sergioadan.dev and the public Jarvis

- A hand-written static site on a Cloudflare Worker; the build fails on any Content-Security-Policy violation or missing file; 14 pull requests through CI, including a security review.
- A public assistant that answers only from the site's pages: hybrid retrieval (BM25 and bge-m3 embeddings, reciprocal rank fusion), streamed answers with citations, Gemma 4 26B on Workers AI's free tier, voice in through Whisper and out through a voice chosen per device.
- Evaluated with fixed questions and two held-out sets committed before the changes they judged: development 16/17, held-out 7/8. Building the evaluation found five real defects.

sergioadan.dev/jarvis · github.com/sadanbermudez/portfolio-site

jarvis-chess: chess commentary that cannot invent tactics

- An MCP server where Stockfish and python-chess compute every fact, and a claim checker marks each statement true, false or uncheckable. On the site, Stockfish 19 runs as WebAssembly in a Web Worker, and move explanations name a tactic only when the engine's own line proves it.
- FIDE Master (peak rating 2402) as the domain judge: every honesty rule fixes a wrong sentence I found in the notes on a real game.

sergioadan.dev/work/chess · github.com/sadanbermudez/jarvis-chess

moe-offload-lab: why a 284B model runs at 0.5 tokens per second

- Modelled bytes per token, benchmarked the SSD (4.8 GB/s, while the engine read 0.72), wrote an AVX-512 FP4 kernel in C++20, profiled the engine and disassembled the shipped binary (no AVX-512 instructions).

- Rebuilt the engine with clang and changed one variable at a time: AVX-512 alone was worth about 9%, while a larger gain of about 55% came from how threads wait at roughly 500 barriers per token on Windows. Reported upstream.

github.com/sadanbermudez/moe-offload-lab

Three Android apps, with working browser demos

- La Piña (Kotlin, Jetpack Compose): a rules engine for a trading method with 263 engine tests. Spanish Tutor (Kotlin, llama.cpp through JNI): an offline tutor on a 2B on-device model. Olympus (Flutter, Riverpod, encrypted SQLite): training and nutrition, with a guard that flags any number from the model that no tool returned.
- Ported each app's logic to JavaScript for the web and checked it against the apps' own test cases; chose the live tutor's model by a 20-turn evaluation (Gemma 4 26B scored 20/20; Qwen3 30B showed wrong Spanish as a correction in 6).

sergioadan.dev/apps

EXPERIENCE

Key Account Manager and Product Specialist

Sensofar-Tech S.L., Barcelona · 2024 – 2026

- Designed an internal LLM-based assistant and a Python document database to centralise technical knowledge and speed up answers from product and scientific documentation.
- Led technical product support and key-account management for optical metrology systems (3D confocal and interferometric) across the UK, Ireland, Italy, Poland, Australia and South-East Europe.
- Ran workshops, demonstrations and training for technical and commercial audiences, including Rolls-Royce, GE Aviation, ST Microelectronics and Red Bull, and turned customer needs into documentation and feedback for product and R&D.

Postdoctoral Researcher, Biomedical Photonics and Sensing

IQS, Universitat Ramon Llull, Barcelona · 2022 – 2024

- EU Horizon 2020 project Light4Lungs: measured singlet oxygen inside bacteria with ultrasensitive photon detection (photodiodes and photomultiplier tubes).
- Designed, built and validated sensing and spectroscopy systems, and coordinated the optics, biochemistry and computational-modelling teams.

Research analysis and postdoc search · Wiesbaden, Germany · 2022, two months: Analysed my PhD results and looked for postdoctoral positions.

EDUCATION

PhD in Engineering, Intelligent Sensing and Measurement · University of Edinburgh · 2017 – 2022

- Thesis: Detection of two-photon photosensitised singlet oxygen in hollow-core photonic crystal fibres. The first quantitative measurement of two-photon singlet oxygen in water, using the fibre as a nanolitre sample cell; seven experiment designs, each changing one thing and measuring again.
- Leadership Award, University of Edinburgh (2019). Talks at the European Society for Photobiology and the Max Planck Institute for the Science of Light.

MSc in Engineering · University of Glasgow · 2016 – 2017 · first year of the Centre for Doctoral Training in Intelligent Sensing and Measurement

MSc in Nanoscience · University of Copenhagen · 2014 – 2016 · photophysics major; machine learning with Python and R

BSc in Biochemistry · Universidad Autónoma de Madrid · 2010 – 2014

SKILLS

AI and LLMs: LLM application engineering, retrieval-augmented generation (RAG): BM25, embeddings, reciprocal rank fusion, AI agents and tool use, Model Context Protocol (MCP), evaluation design: held-out sets, deterministic checks, Inspect AI, local inference: llama.cpp, GGUF quantisation, Vulkan, speech: Whisper, Kokoro, MeloTTS, observability: OpenTelemetry, Arize Phoenix

Software: Python (daily), C++20, JavaScript, WebAssembly and Web Workers, FastAPI and Cloudflare Workers, web security: Content-Security-Policy, Subresource Integrity, profiling and benchmarking, Kotlin, Jetpack Compose, Flutter and Dart (apps built with AI coding agents); Git and GitHub Actions, Docker, Linux and Windows, R, MATLAB

Science and customers: photonics, spectroscopy, experimental design, statistics and curve fitting, instrument control (PyVISA, SCPI) and chess (FIDE Master), key-account management, technical support, workshops and training

LANGUAGES AND PUBLICATIONS

Spanish (native), English (C2), German and French (basic).

S. Adan Bermudez et al., "Designed modular proteins as scaffolds to stabilize fluorescent nanoclusters", *Biomacromolecules* 16, 3836–3844 (2015).